## Next construction: mixed-degree cancellation

Source/literature-derived plan; no theorem or native result claimed.

Select the K=0 prelude and cancellation return in add12_weighted_complete (source lines828–878). Prove mismatch exclusion from valid inputs, plus normalized validity and nonzero weights of the returned degree-one p4. Existing abstract mismatch exclusion assumes equality of squares; this construction derives it from the actual weighted inputs and literal guards.

Write normalized inputs as point $(x, y)$ and $u(U) = U^2 + qU + r$, $v(U) = sU + t$, with four nonzero weights and both Mumford divisibilities against common f. Derive

$$K_{raw} = Z^4 Z_b^4 W\, u(x), \qquad L = Dy, \quad R = Dv(x),$$

where $D = Z^5 Z_b^5 W W_b \neq 0$. Thus K=0 gives shared support, and input validity implies $y^2 = v(x)^2$. This excludes the mismatch None after cancellation fails.

For the exact returned p4, derive $x_4 = -q - x$, $y_4 = v(x_4)$ and $u = (U - x)(U - x_4)$. Its support divides $f - y_4^2$ and its weights $ZZ_b, W_b$ are nonzero. No division by $x - x_4$ or y is needed: retain repeated-support and zero-ordinate cancellation.

**Machine bridge.** Reuse FieldExpr composition under prime $5 \leq p < 2^{31}$ and10 canonical coordinates, preserving source parentheses and guard order. This path needs only existing multiply/square/add/subtract/neg operations. Field.n=1, tables=None and counter headroom remain external. Prove actual cancellation dispatch and canonical valid WordD1 output; independently replay and source-review.

**Controls to freeze.** Nonunit cancellation, repeated support, zero-y precedence, invalid ordinate mismatch, zero weights and p5 boundary. These are algebraic fixtures unless geometric assumptions are separately established.

**Sources.** Ozbay Gurler–Hisil, DOI10.1007/s13389-024-00351-7, sections2,3.3,4.2; exact gate source; prior proof graph and law-remaining-obligations.md. The primary source supplies established mixed-support formulas; no novelty claim. Exact links/hashes and detailed acceptance targets are in PLAN.md/status.json.

Next after this slice: generic K-nonzero output congruence/divisibility, then positive shared-support composition using reviewed doubling. Tripling remains separate. No cohort or cost scope is added.